$$
\begin{array}{rcl}
\mathit{intexp} & ::= & \mathit{nat} \mid \mathit{var} \mid -_u\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ +\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ -_b\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \times\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \div\ \mathit{intexp} \\
\mathit{boolexp} & ::= & \textbf{true} \mid \textbf{false} \\
& \mid & \mathit{intexp}\ =\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ <\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ >\ \mathit{intexp} \\
& \mid & \mathit{boolexp} \wedge \mathit{boolexp} \\
& \mid & \mathit{boolexp} \vee \mathit{boolexp} \\
& \mid & \neg\ \mathit{boolexp} \\
\mathit{comm} & ::= & \textbf{skip} \\
& \mid & \mathit{var} := \mathit{intexp} \\
& \mid & \mathit{comm}; \mathit{comm} \\
& \mid & \textbf{if}\ \mathit{boolexp}\ \textbf{then}\ \mathit{comm}\ \textbf{else}\ \mathit{comm} \\
& \mid & \textbf{repeat}\ \mathit{comm}\ \mathit{boolexp}
\end{array}
$$

$$
\begin{array}{rcl}
\mathit{digit} & ::= & \texttt{'0'} \mid \texttt{'1'} \mid \ldots \mid \texttt{'9'} \\
\mathit{letter} & ::= & \texttt{'a'} \mid \ldots \mid \texttt{'Z'} \\
\mathit{nat} & ::= & \mathit{digit} \mid \mathit{digit}\ \mathit{nat} \\
\mathit{var} & ::= & \mathit{letter} \mid \mathit{letter}\ \mathit{var}
\end{array}
$$

$$
\begin{array}{rcl}
\mathit{intexp} & ::= & \mathit{nat} \\
& \mid & \mathit{var} \\
& \mid & \texttt{'-'}\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \texttt{'+'}\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \texttt{'-'}\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \texttt{'*'}\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \texttt{'/'}\ \mathit{intexp} \\
& \mid & \texttt{'('}\ \mathit{intexp}\ \texttt{')'} \\
\mathit{boolexp} & ::= & \texttt{'true'} \mid \texttt{'false'} \\
& \mid & \mathit{intexp}\ \texttt{'='}\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \texttt{'<'}\ \mathit{intexp} \\
& \mid & \mathit{intexp}\ \texttt{'>'}\ \mathit{intexp} \\
& \mid & \mathit{boolexp}\ \texttt{'\&'}\ \mathit{boolexp} \\
& \mid & \mathit{boolexp}\ \texttt{'|'}\ \mathit{boolexp} \\
& \mid & \texttt{'\~{}'}\ \mathit{boolexp} \\
& \mid & \texttt{'('}\ \mathit{boolexp}\ \texttt{')'} \\
\mathit{comm} & ::= & \texttt{'skip'} \\
& \mid & \mathit{var}\ \texttt{':='}\ \mathit{intexp} \\
& \mid & \mathit{comm}\ \texttt{';'}\ \mathit{comm} \\
& \mid & \texttt{'if'}\ \mathit{boolexp}\ \texttt{'then'}\ \mathit{comm}\ \texttt{'else'}\ \mathit{comm}\ \texttt{'end'} \\
& \mid & \texttt{'repeat'}\ \mathit{comm}\ \texttt{'until'}\ \mathit{boolexp}\ \texttt{'end'}
\end{array}
$$

$$\frac{}{\langle nv, \sigma\rangle \Downarrow_{\text{intexp}} nv}\ \textsc{NVal} \qquad \frac{}{\langle v, \sigma\rangle \Downarrow_{\text{intexp}} \sigma v}\ \textsc{Var} \qquad \frac{\langle e, \sigma\rangle \Downarrow_{\text{intexp}} n}{\langle -_u e, \sigma\rangle \Downarrow_{\text{intexp}} -n}\ \textsc{UMinus}$$

$$\frac{\langle e_0, \sigma\rangle \Downarrow_{\text{intexp}} n_0 \quad \langle e_1, \sigma\rangle \Downarrow_{\text{intexp}} n_1}{\langle e_0 + e_1, \sigma\rangle \Downarrow_{\text{intexp}} n_0 + n_1}\ \textsc{Plus} \qquad \text{(análogamente para } -_b,\ \times\text{)}$$

$$\frac{\langle e_0, \sigma\rangle \Downarrow_{\text{intexp}} n_0 \quad \langle e_1, \sigma\rangle \Downarrow_{\text{intexp}} n_1 \quad n_1 \neq 0}{\langle e_0 \div e_1, \sigma\rangle \Downarrow_{\text{intexp}} n_0 \div n_1}\ \textsc{Div}$$

$$\frac{\langle e_0, \sigma\rangle \Downarrow_{\text{intexp}} n_0 \quad \langle e_1, \sigma\rangle \Downarrow_{\text{intexp}} n_1}{\langle e_0 = e_1, \sigma\rangle \Downarrow_{\text{boolexp}} n_0 = n_1}\ \textsc{Eq} \qquad \text{(análogamente para } <\text{ y } >\text{)}$$

$$\frac{}{\langle bv, \sigma\rangle \Downarrow_{\text{boolexp}} bv}\ \textsc{BVal} \qquad \frac{\langle p, \sigma\rangle \Downarrow_{\text{boolexp}} b}{\langle \neg p, \sigma\rangle \Downarrow_{\text{boolexp}} \neg b}\ \textsc{Not}$$

$$\frac{\langle p_0, \sigma\rangle \Downarrow_{\text{boolexp}} b_0 \quad \langle p_1, \sigma\rangle \Downarrow_{\text{boolexp}} b_1}{\langle p_0 \vee p_1, \sigma\rangle \Downarrow_{\text{boolexp}} b_0 \vee b_1}\ \textsc{Or} \qquad \text{(análogamente para } \wedge\text{)}$$

$$\frac{\langle e, \sigma\rangle \Downarrow_{\text{intexp}} n}{\langle v := e, \sigma\rangle \rightsquigarrow \langle \textsf{skip}, [\,\sigma \mid v{:}\, n\,]\rangle}\ \textsc{Ass}$$

$$\frac{}{\langle \textsf{skip}; c_1, \sigma\rangle \rightsquigarrow \langle c_1, \sigma\rangle}\ \textsc{Seq}_1 \qquad \frac{\langle c_0, \sigma\rangle \rightsquigarrow \langle c_0', \sigma'\rangle}{\langle c_0; c_1, \sigma\rangle \rightsquigarrow \langle c_0'; c_1, \sigma'\rangle}\ \textsc{Seq}_2$$

$$\frac{\langle b, \sigma\rangle \Downarrow_{\text{boolexp}} \textsf{true}}{\langle \textsf{if}\ b\ \textsf{then}\ c_0\ \textsf{else}\ c_1, \sigma\rangle \rightsquigarrow \langle c_0, \sigma\rangle}\ \textsc{If}_1 \qquad \frac{\langle b, \sigma\rangle \Downarrow_{\text{boolexp}} \textsf{false}}{\langle \textsf{if}\ b\ \textsf{then}\ c_0\ \textsf{else}\ c_1, \sigma\rangle \rightsquigarrow \langle c_1, \sigma\rangle}\ \textsc{If}_2$$

$$\frac{}{\langle \textsf{repeat}\ c\ \textsf{until}\ b, \sigma\rangle \rightsquigarrow \langle c; \textsf{if}\ b\ \textsf{then skip else repeat}\ b\ \textsf{until}\ c, \sigma\rangle}\ \textsc{Repeat}$$